

Multiple Access

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Figure *Data link layer divided into two functionality-oriented sublayers*

Data link layer

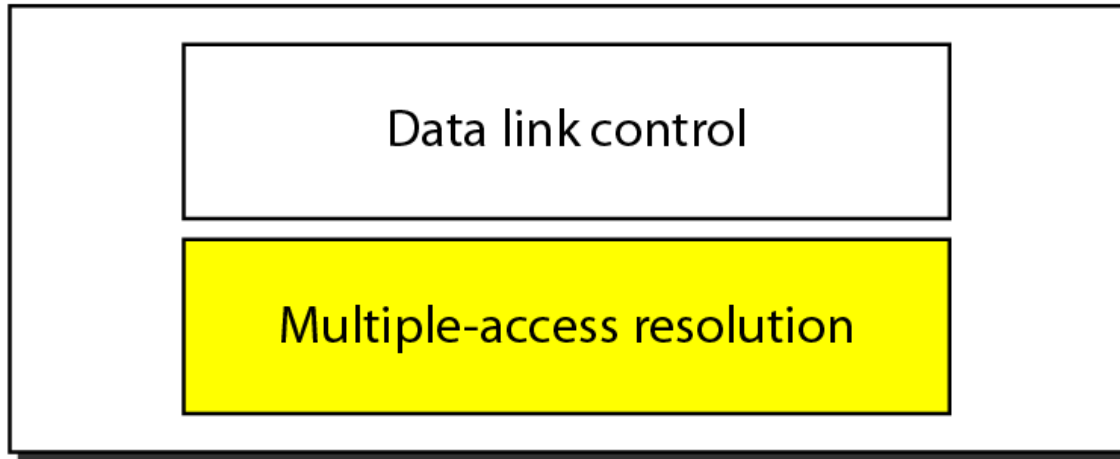
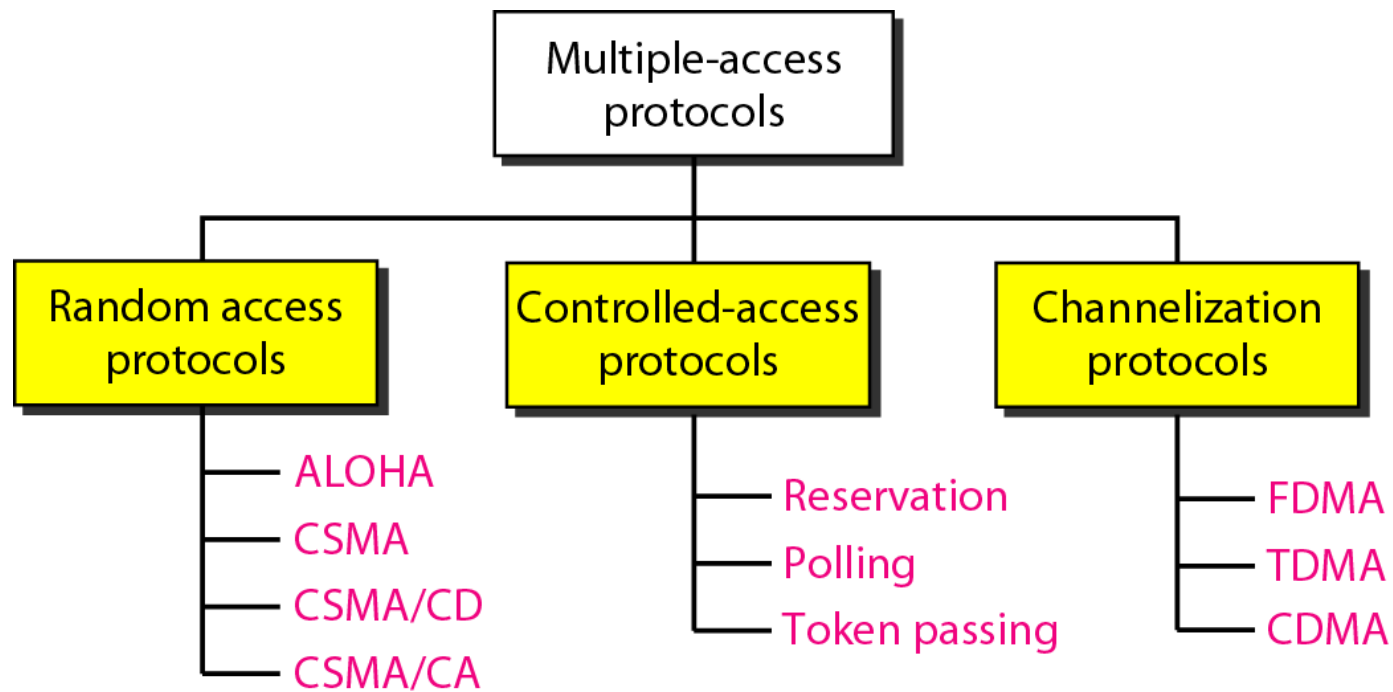


Figure *Taxonomy of multiple-access protocols discussed in this chapter*



RANDOM ACCESS

*In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another.*

No station permits, or does not permit, another station to send.

At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.

Topics discussed in this section:

ALOHA

Carrier Sense Multiple Access

Carrier Sense Multiple Access with Collision Detection

Carrier Sense Multiple Access with Collision Avoidance

Random Access

- In **Random Access Protocols**, all nodes have equal priority.
- Thus, each station has the right to the medium without being controlled by any other station.
- No fixed schedule
- Any station can transmit anytime
- Collisions may occur
- So these protocols are also called:
“Contention-Based Protocols”

Key Features of Random Access

- Decentralized control
- Suitable for bursty traffic
- Simple implementation
- Collision possible
- Retransmission needed

Types of Random Access Protocols

- **ALOHA**
- **Slotted ALOHA**
- **CSMA**
- **CSMA/CD**
- **CSMA/CA**

1. Pure ALOHA Protocol

- Pure ALOHA is the **simplest random access protocol.**
- A node transmits whenever it has data.

Working

- Station sends frame immediately
- If collision occurs, **frame is destroyed**
- Station waits random time and retransmits

Example

Suppose:

- Node A sends at time t
- Node B sends at time $t + \text{small delay}$
- Frames overlap $\rightarrow$ collision occurs $\rightarrow$ both retransmit later

Vulnerable Time

- **Vulnerable time is the time interval during which transmission of a frame is susceptible to collision if any other station transmits.**

Vulnerable Time

- **Vulnerable time** is the time interval during which a transmitted frame is **susceptible to collision**.
- In simple words:
 - If any other station transmits during this interval, the frame will overlap and a **collision** will occur.
- So, vulnerable time is the **danger period** for a frame.

Vulnerable Time

When Does Collision Occur?

Suppose Station A starts transmitting at time t .

- Its frame occupies the channel from:
 - $t \rightarrow t + T$
- Now collision will happen if another station starts transmitting:

1. Within T seconds before time t

Because its frame will still be on the channel when A starts.

2. Within T seconds after time t

- Because it will overlap with A's ongoing frame.

Vulnerable Time

- **Total Vulnerable Interval**

- So collision is possible during:

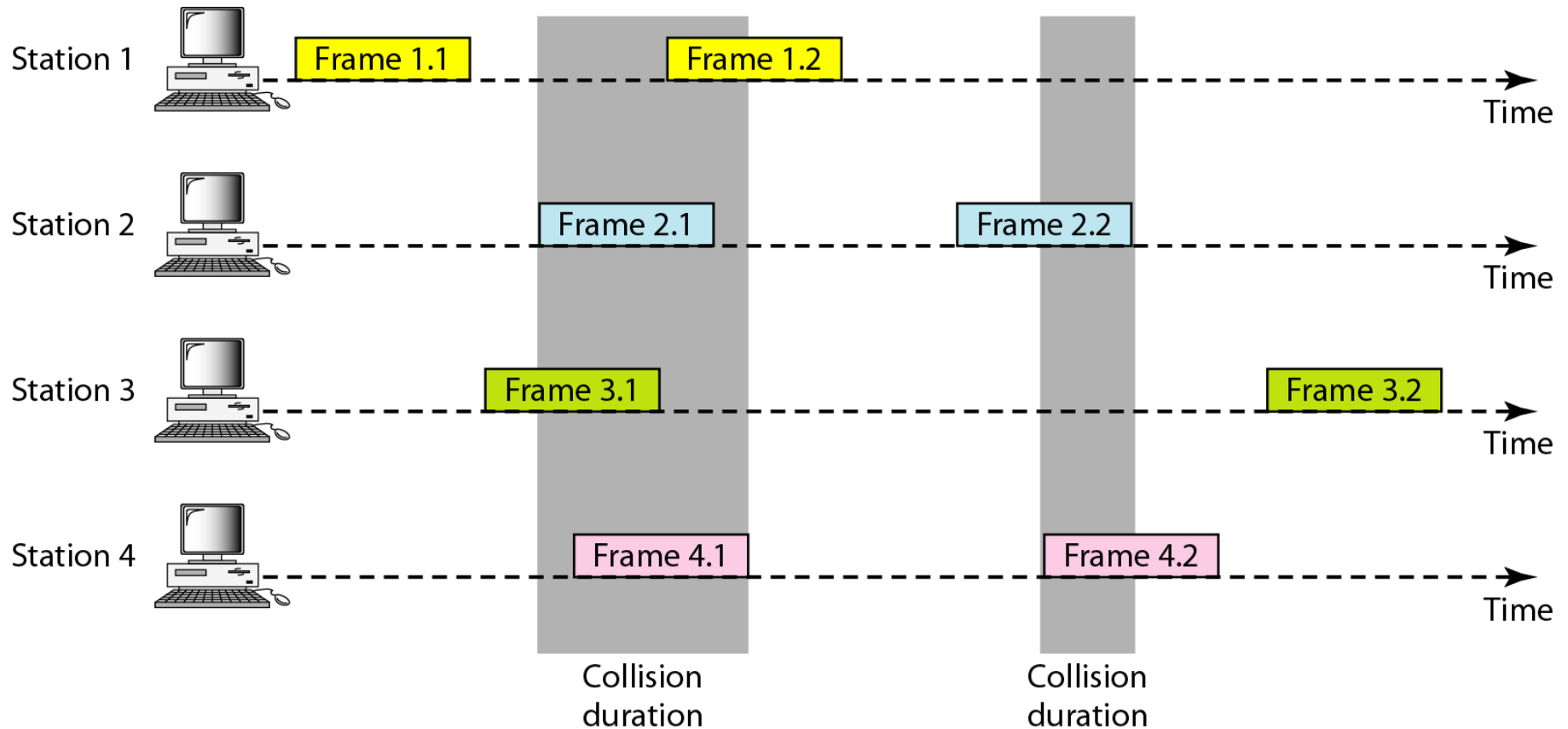
$$T \text{ (before)} + T \text{ (after)} = 2T$$

- Therefore,

Vulnerable Time in Pure ALOHA:

$$2T$$

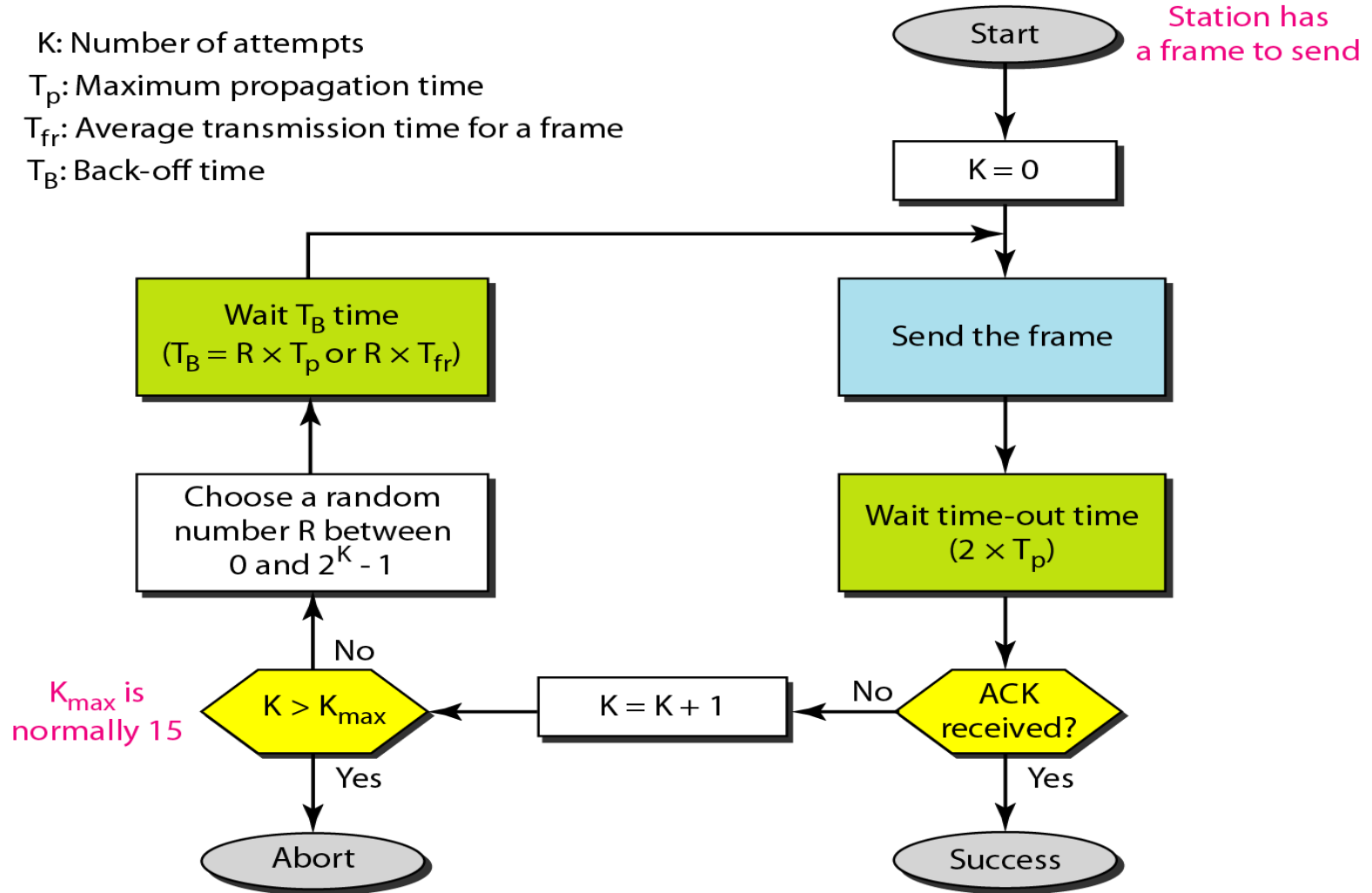
Figure *Frames in a pure ALOHA network*



Frames in a pure ALOHA network

- The pure ALOHA protocol relies on acknowledgments from the receiver.
 - When a station sends a frame, it **expects the receiver to send an acknowledgment.**
 - If the **acknowledgment does not arrive after a time-out period**, the station assumes that the **frame (or the acknowledgment)** has been destroyed and resends the frame.
-

Figure *Procedure for pure ALOHA protocol*



Note

The throughput for pure ALOHA is

$$S = G \times e^{-2G} .$$

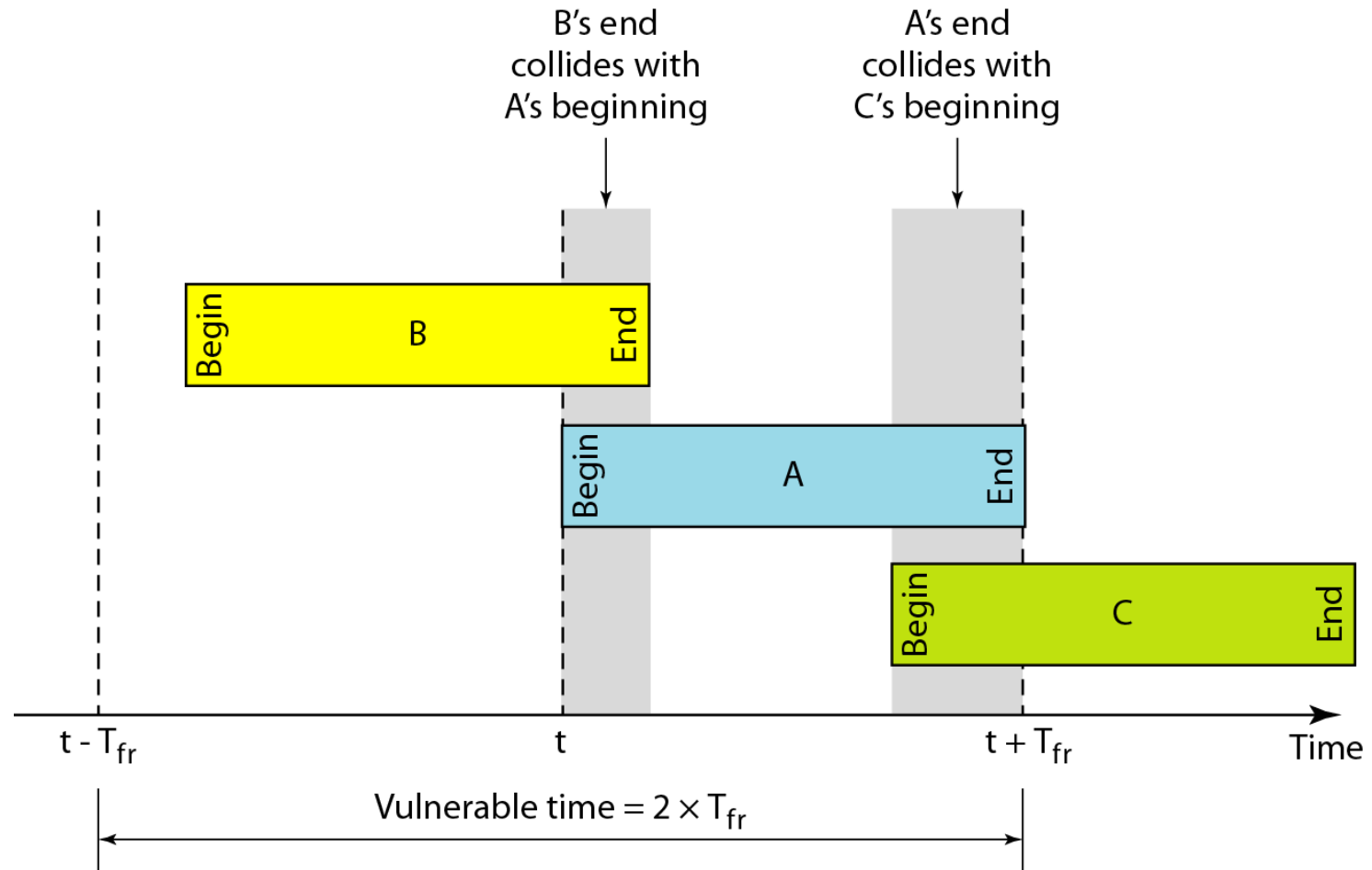
The maximum throughput

$$S_{\max} = 0.184(18.4\%) \text{ when } G = (1/2).$$

Where:

- *S = throughput*
- *G = offered load*

Figure *Vulnerable time for pure ALOHA protocol*



Example 3

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

- a. 1000 frames per second b. 500 frames per second*
- c. 250 frames per second.*

Solution

The frame transmission time is 200/200 kbps or 1 ms.

- a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1. In this case $S = G \times e^{-2G}$ or $S = 0.135$ (13.5 percent). This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.*

Example 3 (continued)

- b. If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2). In this case $S = G \times e^{-2G}$ or $S = 0.184$ (18.4 percent). This means that the throughput is $500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive. Note that this is the maximum throughput case, percentage-wise.*
- c. If the system creates 250 frames per second, this is (1/4) frame per millisecond. The load is (1/4). In this case $S = G \times e^{-2G}$ or $S = 0.152$ (15.2 percent). This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive.*

2. Slotted ALOHA Protocol

- Slotted ALOHA improves Pure ALOHA by **dividing time into slots.**
- Transmission allowed only at slot boundaries.

Working

- Time divided into slots of length T
- Node transmits only at start of slot
- Collision occurs only if two nodes transmit in same slot

Slotted Aloha

Example

- Slot 1: Node A transmits → success
- Slot 2: Node B and Node C transmit → collision

Vulnerable Time in Slotted ALOHA

Slotted ALOHA Rule

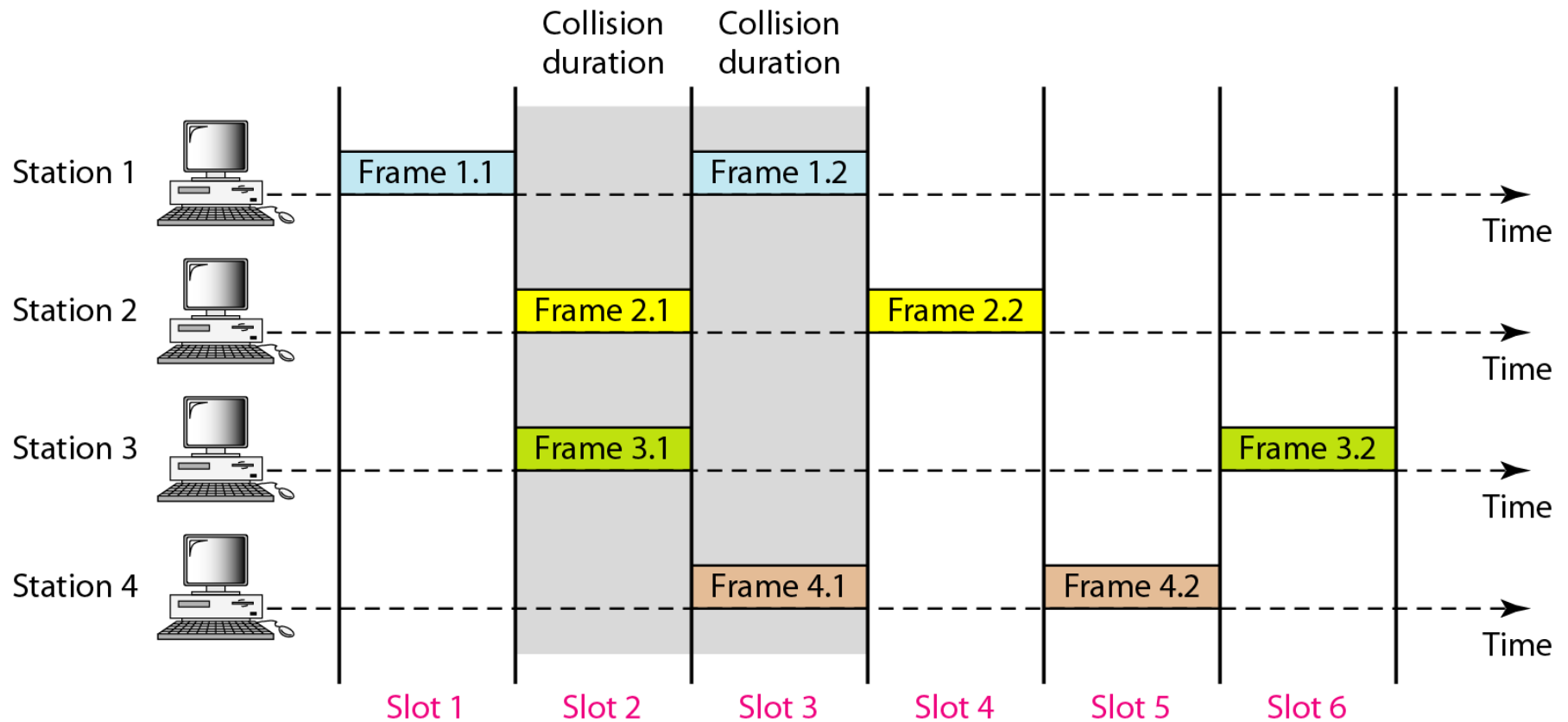
- Transmission happens only at the **start of time slots**.
- So overlap is possible only inside the same slot.

Collision Condition

- Collision occurs only if another node transmits in the **same slot**.
- So vulnerable time becomes:

■ T

Figure 6 *Frames in a slotted ALOHA network*





Note

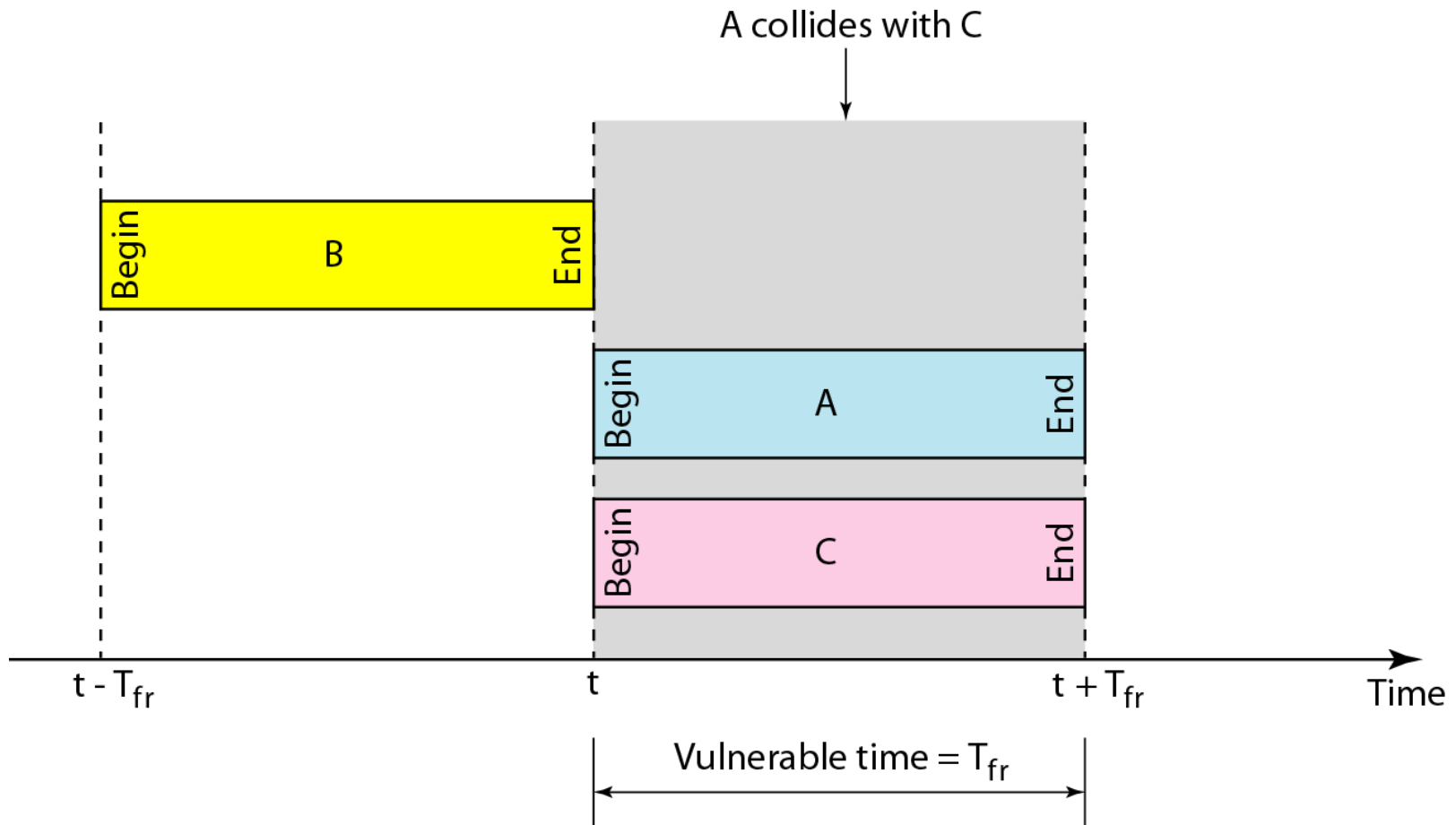
The throughput for slotted ALOHA is

$$S = G \times e^{-G} .$$

The maximum throughput

$$S_{\max} = 0.368 \text{ when } G = 1.$$

Figure 7 *Vulnerable time for slotted ALOHA protocol*



Example 4

A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

- a. 1000 frames per second b. 500 frames per second*
- c. 250 frames per second.*

Solution

The frame transmission time is 200/200 kbps or 1 ms.

- a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1. In this case $S = G \times e^{-G}$ or $S = 0.368$ (36.8 percent). This means that the throughput is $1000 \times 0.0368 = 368$ frames. Only 386 frames out of 1000 will probably survive.*

Example 4 (continued)

- b. If the system creates 500 frames per second, this is $(1/2)$ frame per millisecond. The load is $(1/2)$. In this case $S = G \times e^{-G}$ or $S = 0.303$ (30.3 percent). This means that the throughput is $500 \times 0.303 = 151$. Only 151 frames out of 500 will probably survive.*
- c. If the system creates 250 frames per second, this is $(1/4)$ frame per millisecond. The load is $(1/4)$. In this case $S = G \times e^{-G}$ or $S = 0.195$ (19.5 percent). This means that the throughput is $250 \times 0.195 = 49$. Only 49 frames out of 250 will probably survive.*

CSMA-Carrier Sense Multiple Access

- It is a **random access (contention-based)** Medium Access Control protocol used in shared networks.
- CSMA was developed to improve the performance of ALOHA by reducing collisions.
- The main principle of CSMA is:
“Listen before you transmit.”
- Before sending data, a station first checks whether the communication channel is free or busy.
- CSMA is widely used in networking technologies like Ethernet and Wi-Fi

Why CSMA is Needed?

In ALOHA:

- Stations transmit anytime
- Collisions are very frequent
- Channel efficiency is low
- CSMA reduces collisions by sensing the channel first.

Working of CSMA Protocol

The working steps are:

- A station has a frame to send.
- It senses (listens to) the channel.
- If the channel is idle ,it transmits immediately.
- If the channel is busy, it waits according to the CSMA rule.
- Collisions may still occur due to propagation delay.

Example of CSMA

Suppose:

- Node A wants to transmit.
- Node B also wants to transmit.

Case 1: Channel Idle

- Node A senses channel free and sends frame.
- Node B senses channel busy and waits.
- So **collision is avoided**.

Case 2: Collision Due to Propagation Delay

- Node A starts transmission.
- Before signal reaches Node B, Node B also senses channel idle.
- Node B transmits.
- Result: **Collision occurs**.

Figure 8 *Space/time model of the collision in CSMA*

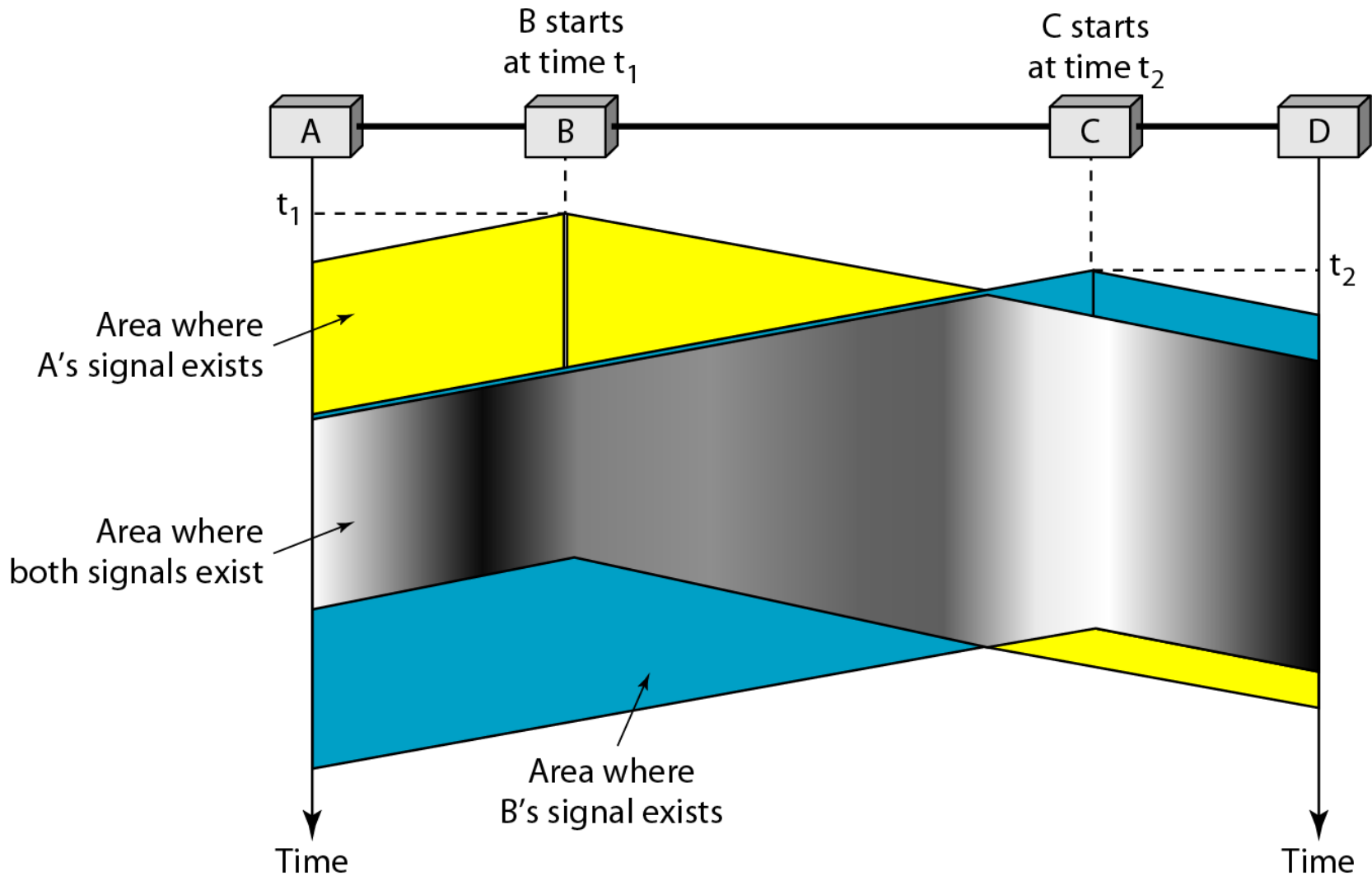
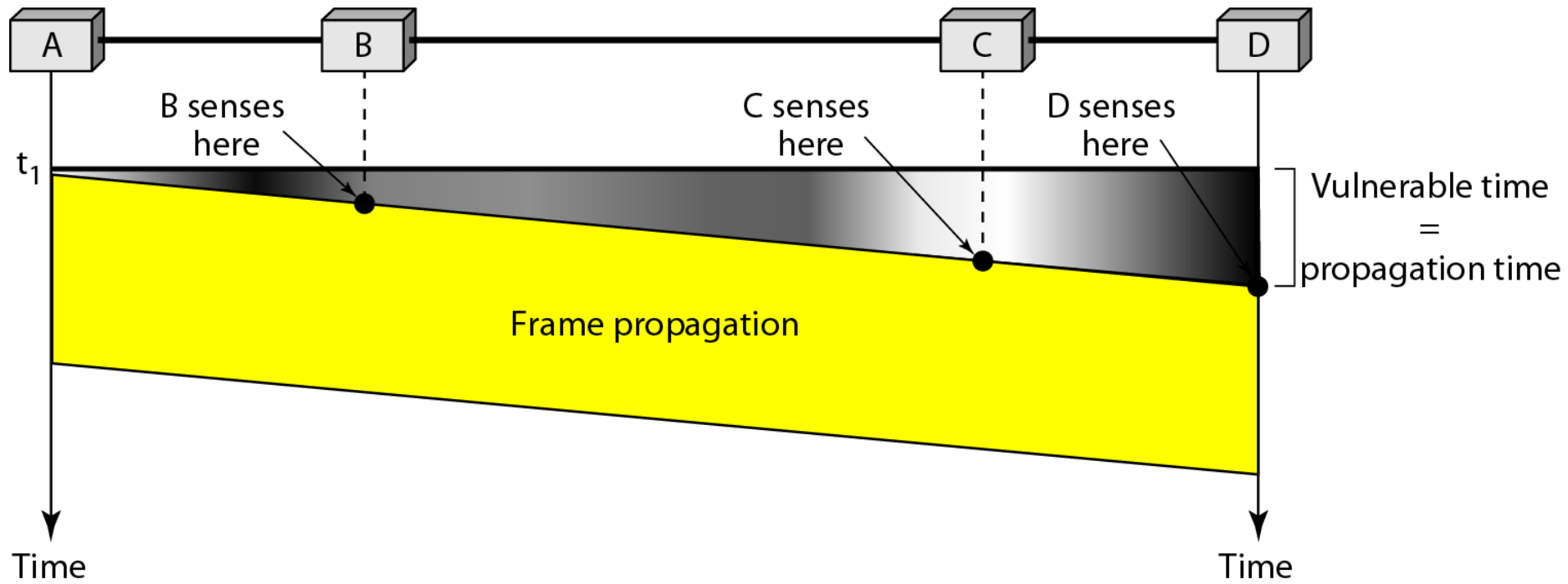


Figure 9 *Vulnerable time in CSMA*



1-Persistent CSMA

- Station **continuously senses** the medium.
- If medium is **free** → transmit immediately.
- If medium is **busy** → keep sensing until free, then transmit.

Pros:

- Quick transmission once medium is free.

Cons:

- High chance of collision if multiple stations are waiting.

Example Scenario:

- **Two computers (A and B)** want to send data.
- Both sense the medium and if they find it free, both send at the same time.
Therefore, a collision occurs.
- If Collision detected, then wait and retry after random backoff.

Non-Persistent CSMA

Station senses the medium.

- If medium is **free**, transmit immediately.
- If medium is **busy**, wait for a **random time**, then sense again.

Pros:

- Reduces collisions compared to 1-persistent.

Cons:

- Transmission **delays increase**, especially under light traffic.

Example Scenario:

- A, B, C want to send data.
- A senses busy, then waits random time, then senses again.
- B senses medium free, then it transmits first.
- A senses again, transmits after B is done and thus collision avoided.

P-Persistent CSMA (used in slotted channels)

- Time is divided into **slots**.
- When medium is free:
 - Transmit with **probability p** .
 - Wait for next slot with probability **$(1-p)$** .
- If medium is busy, then wait next slot and repeat.

Pros:

- Reduces collisions compared to 1-persistent.
- Works well in **moderate to high traffic**.

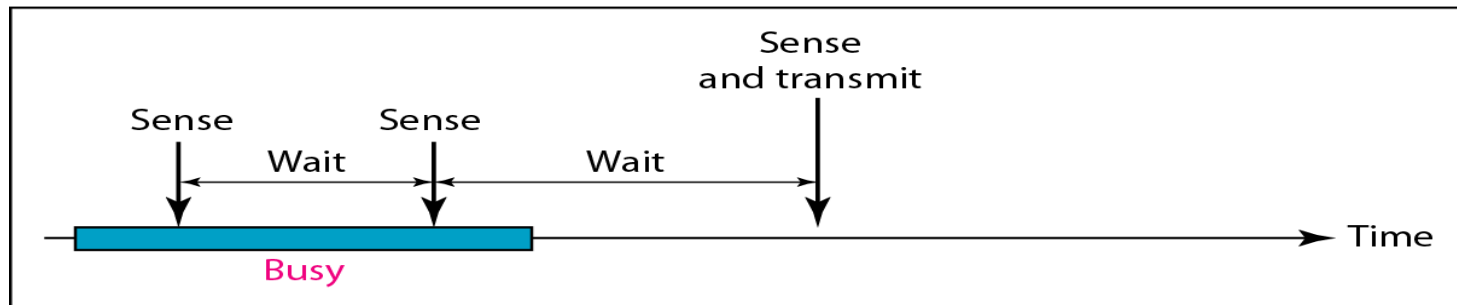
Example Scenario:

- Probability $p = 0.5$, two stations A and B waiting.
- Time slot 1: A decides to transmit (50%), B decides not to (50%), Therefore, A transmits, B waits.
- Time slot 2: B senses again, now it may transmit.

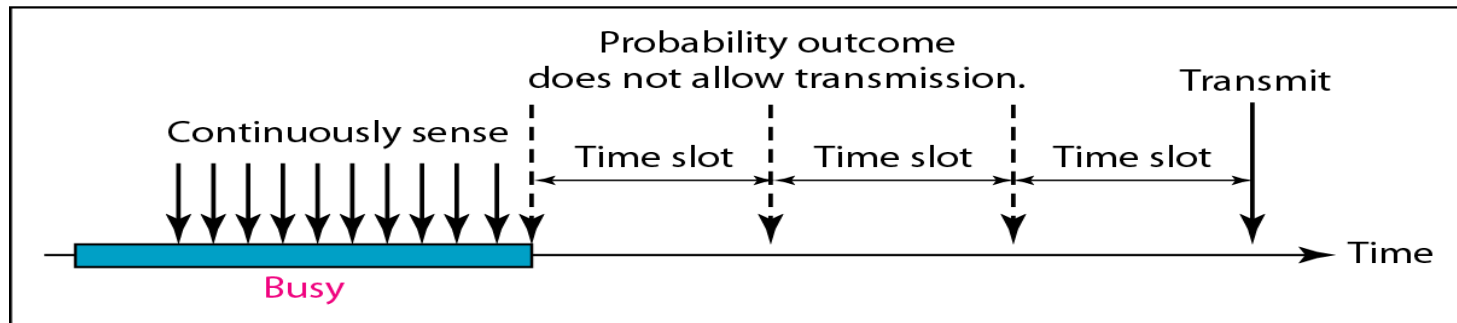
Figure 10 *Behavior of three persistence methods*



a. 1-persistent

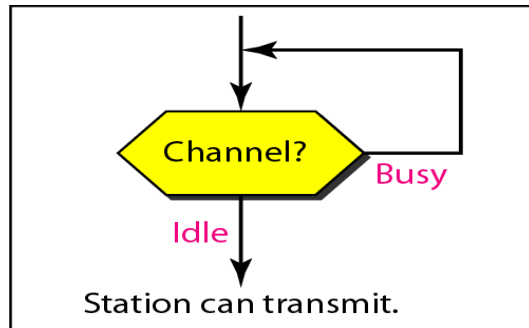


b. Nonpersistent

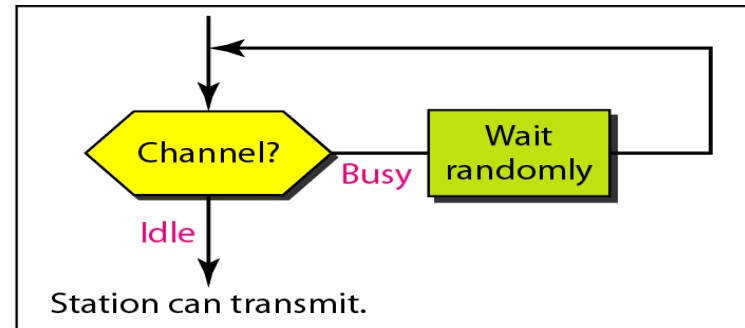


c. p-persistent

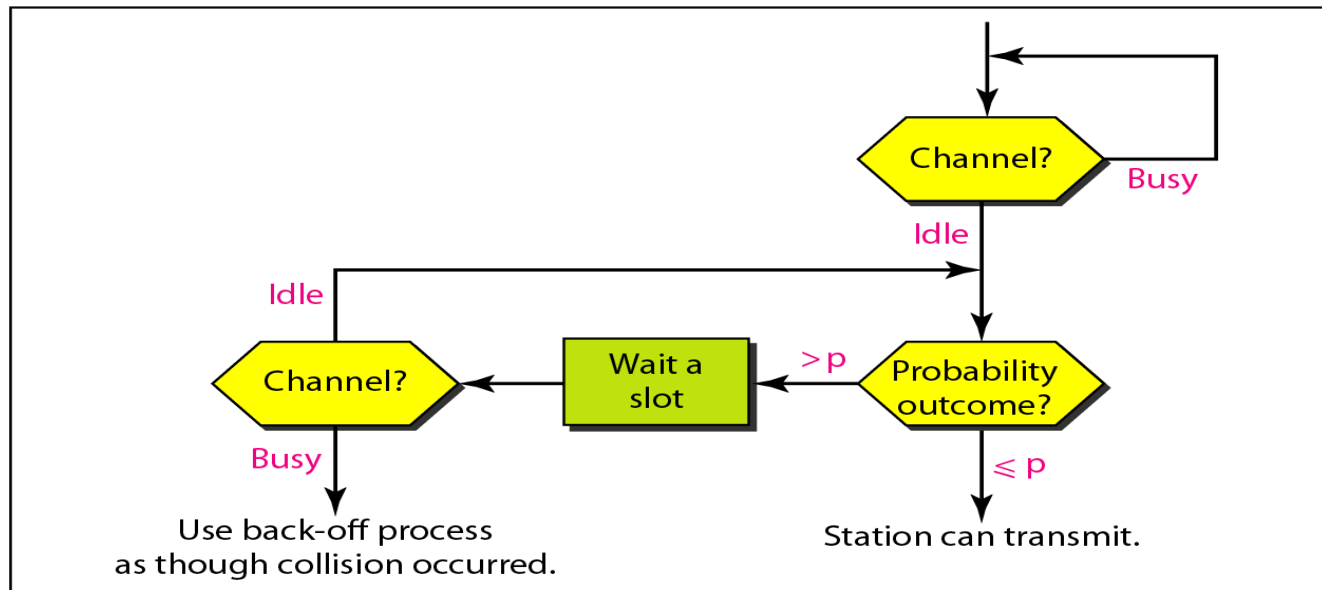
Figure 11 *Flow diagram for three persistence methods*



a. 1-persistent



b. Nonpersistent



c. p-persistent

CSMA/CD

- **CSMA/CD** stands for **Carrier Sense Multiple Access with Collision Detection**.
- It is a network protocol used to **control how multiple devices share a communication channel** so that they don't transmit at the same time.
- It was widely used in traditional **Ethernet LANs with shared media (like coaxial cable)**.

Working of CSMA/CD

1. Carrier Sense (Listen First)

- Before sending data, a device *listens to the network cable* to check if someone else is transmitting.
- If the cable is *busy*, it waits.
- If the cable is *quiet*, it proceeds.

2. Multiple Access

- All devices (computers, stations) on the same network segment share the *same cable*.
- Everyone has *equal opportunity* to transmit when the cable is free.

3. Collision Detection

- While transmitting, the device still monitors the network.
- If it senses a collision (two devices sent at the same time), it stops sending immediately.
- It then sends a jam signal to let all other devices know a collision happened.

4. Backoff and Retry

- After a collision, each affected device waits for a random backoff time before trying again.
- This random delay reduces the chance of repeated collisions.

EXAMPLE

- Two computers **A** and **B** share the same cable.
- Both want to send data at the same time.
- **How CSMA/CD Works?**
 - **A and B both listen** — they find the cable idle.
 - Both start transmitting at **the same time**.
 - Because both signals go onto the cable simultaneously, **their signals collide**.
 - The collision causes garbled data.
 - Both computers **detect the collision** (by sensing that the signal on the cable does not match what they sent).
 - Both **stop transmitting**, send a **jam signal**, then **wait a random time** before trying again.
 - This mechanism ensures that though collisions can't be *prevented*, they are *detected quickly* and resolved fairly.

Vulnerable Time

- There is a short period during which two devices might think the channel is free and start transmitting simultaneously — leading to a collision.
- This period equals the *time it takes for a signal to travel across the network* (propagation time).

Minimum Frame Size

- To ensure that a sender can detect a collision, the **frame must be long enough** so that the signal is still being transmitted when a collision is detected.
- Mathematically:
Frame transmission time $\geq 2 \times$ propagation time
- If frames are too short, a collision might occur after transmission finishes — and won't be detected.

What should be the Minimum length of the Packet to be Transmitted?

- Transmission Time = T_t = Length of the packet/ Bandwidth of the link.
- Substituting above, we get,
- Length of the packet/ Bandwidth of the link $\geq 2 * T_p$
- **Length of the packet $\geq 2 * T_p * \text{Bandwidth of the link}$**

Key Features of CSMA/CD

- **Carrier Sense:** Devices listen before sending.
- **Shared Medium:** All devices share the same cable.
- **Collision Detection:** Collisions are actively detected.
- **Jam Signal:** Used to notify others about a collision.
- **Random Backoff:** Devices wait random times to retry.
- **Minimum Frame Size:** Ensures collisions can be caught.
- CSMA/CD works only when communication is **half-duplex**, meaning:

A device can **either send or receive** at one time (not both). So collisions are possible, thus CSMA/CD is needed.

Limitations of CSMA/CD

- Not very scalable — performance drops with many devices.
- Collisions still occur and reduce throughput.
- Not used in modern switched, full-duplex Ethernet networks.

Figure 12 *Collision of the first bit in CSMA/CD*

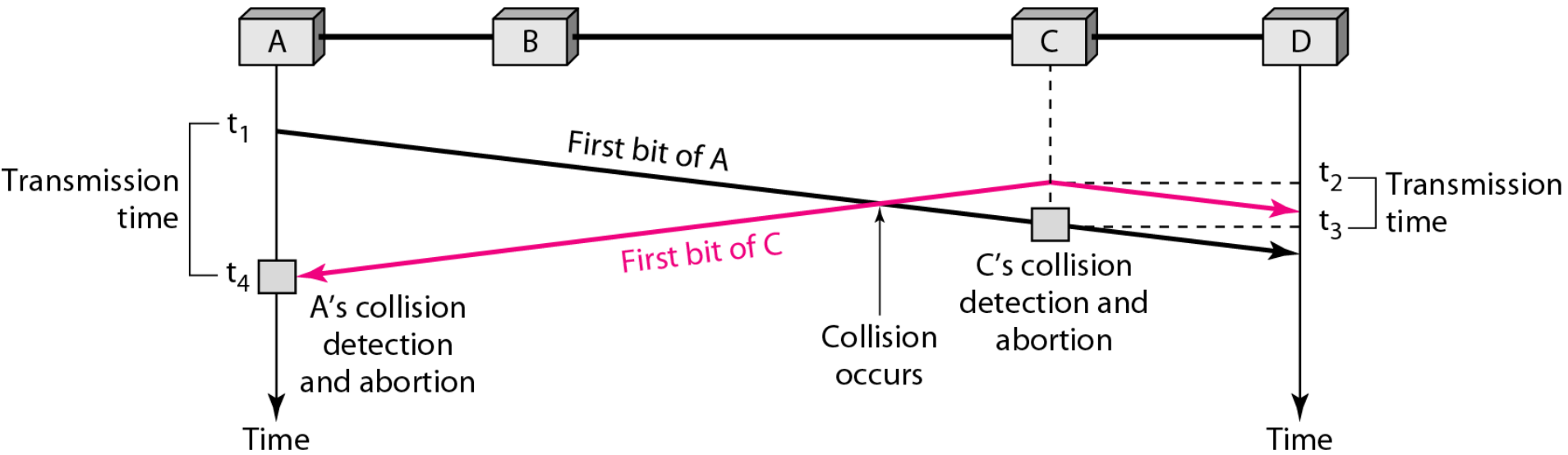
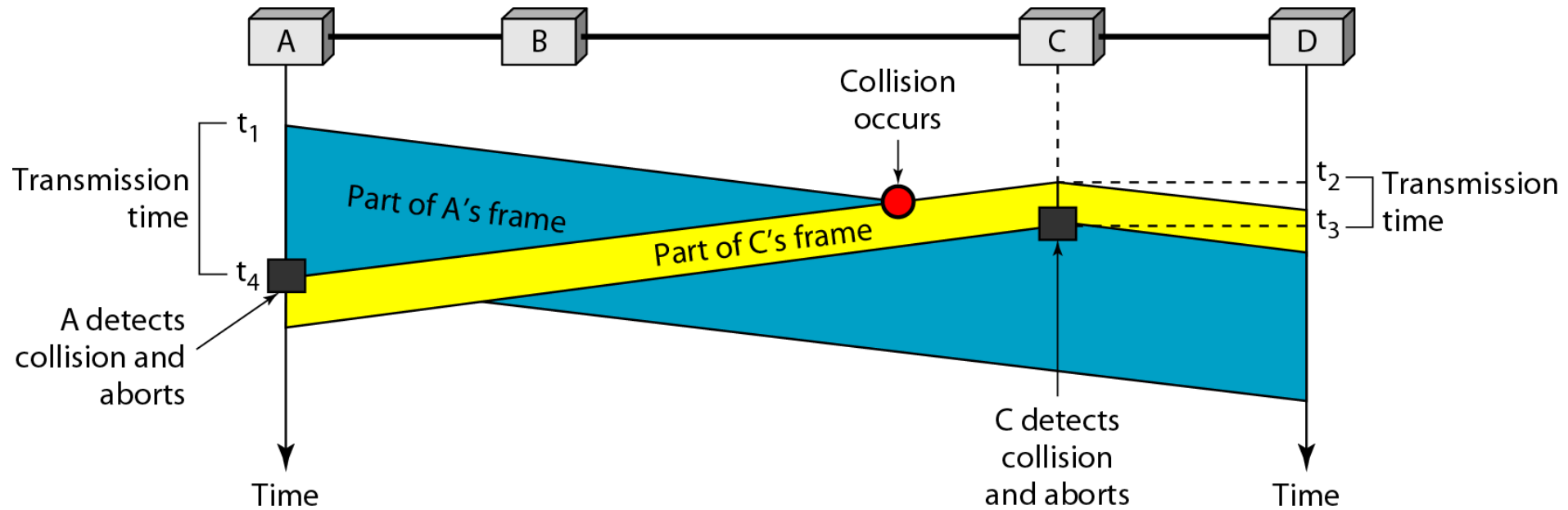


Figure 13 *Collision and abortion in CSMA/CD*



Minimum Frame Size

- For *CSMA/CD* to work, we need a restriction on the frame size. Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission.
 - This is so because the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection. Therefore, the frame transmission time T_{fr} must be at least two times the maximum propagation time T_p .
 - To understand the reason, let us think about the worst-case scenario. If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time T_p to reach the second, and the effect of the collision takes another time T_p to reach the first.
 - **The requirement is that the first station must still be transmitting after $2T_p$.**
-



Example 5

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is $25.6 \mu\text{s}$, what is the minimum size of the frame?

Solution

The frame transmission time is $T_{fr} = 2 \times T_p = 51.2 \mu\text{s}$. This means, in the worst case, a station needs to transmit for a period of $51.2 \mu\text{s}$ to detect the collision. The minimum size of the frame is $10 \text{ Mbps} \times 51.2 \mu\text{s} = 512$ bits or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet.

Figure 14 *Flow diagram for the CSMA/CD*

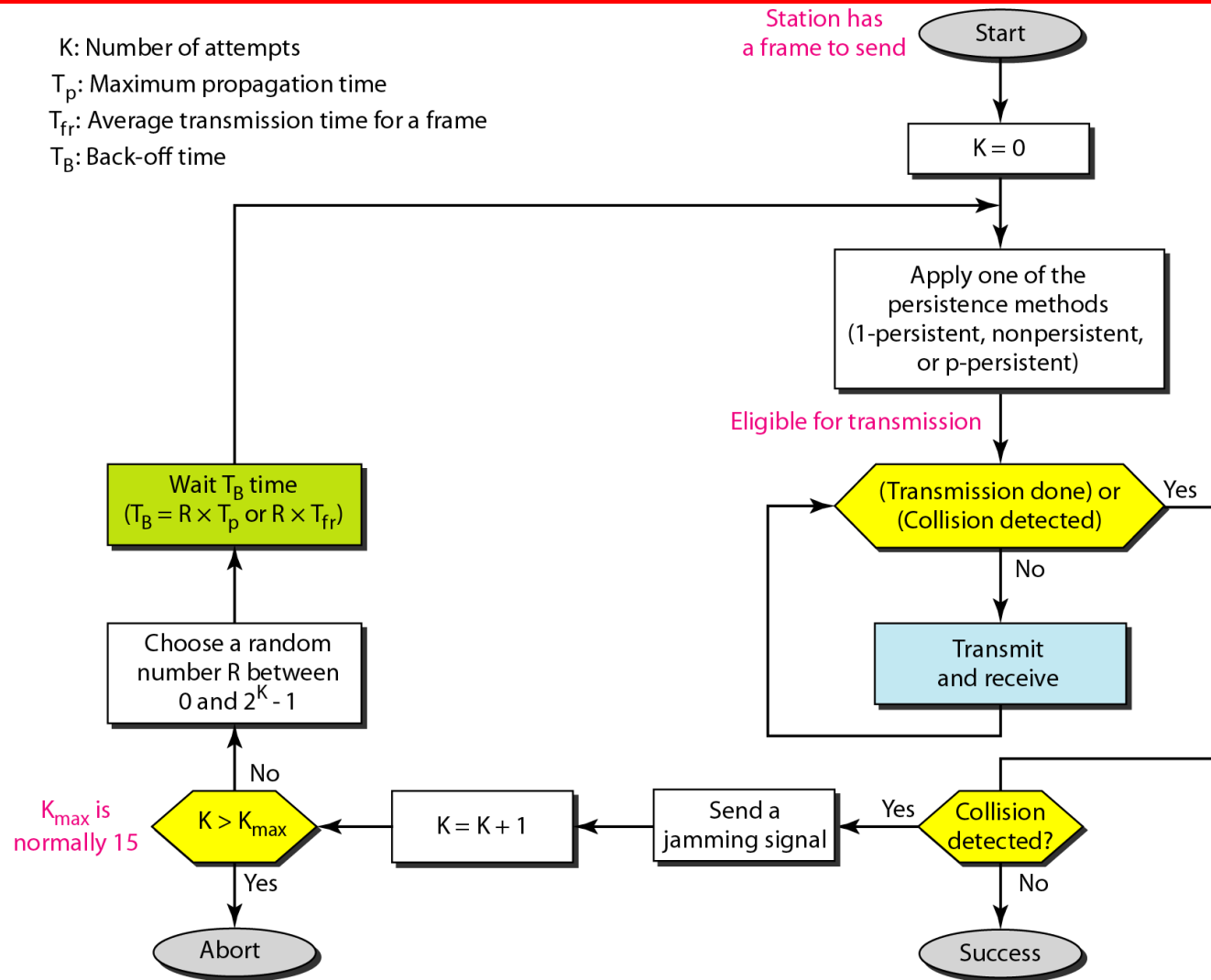
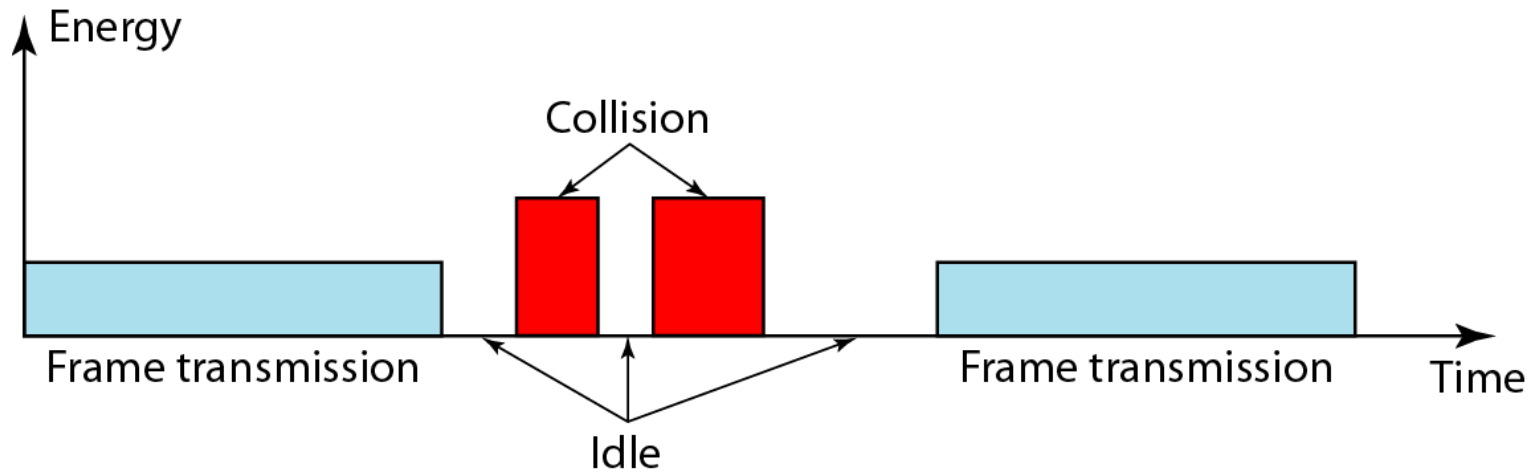


Figure 15 *Energy level during transmission, idleness, or collision*



CSMA-CA

- CSMA/CA stands for **Carrier Sense Multiple Access with Collision Avoidance**, a protocol used mainly in wireless networks like Wi-Fi (IEEE 802.11) to manage shared channel access and prevent data collisions

Why CSMA/CA is Needed?

Problem in Wireless Networks

- In wired Ethernet networks, we use **CSMA/CD** (Collision Detection) because devices can detect collisions while transmitting.
- However, in **wireless networks**, collision detection is not practical.

Reasons:

- A wireless device cannot listen to the channel while transmitting at the same time.
- The transmitted signal becomes much stronger than any incoming signal, so collision detection is difficult.

Why CSMA/CA is Needed?

- Wireless networks also suffer from the **hidden terminal problem**, where two devices may not hear each other but still transmit to the same receiver.
- Therefore, instead of detecting collisions after they occur, wireless networks use:
CSMA/CA (Collision Avoidance) —which tries to prevent collisions before transmission.

How It Works

- Devices "listen" to the channel before transmitting; if busy, they wait a random backoff time before retrying, reducing collision odds.
- An **acknowledgment (ACK)** from the receiver confirms successful delivery, prompting retransmission if absent.
- This contrasts with CSMA/CD (used in wired Ethernet), which detects collisions after they occur rather than avoiding them upfront.

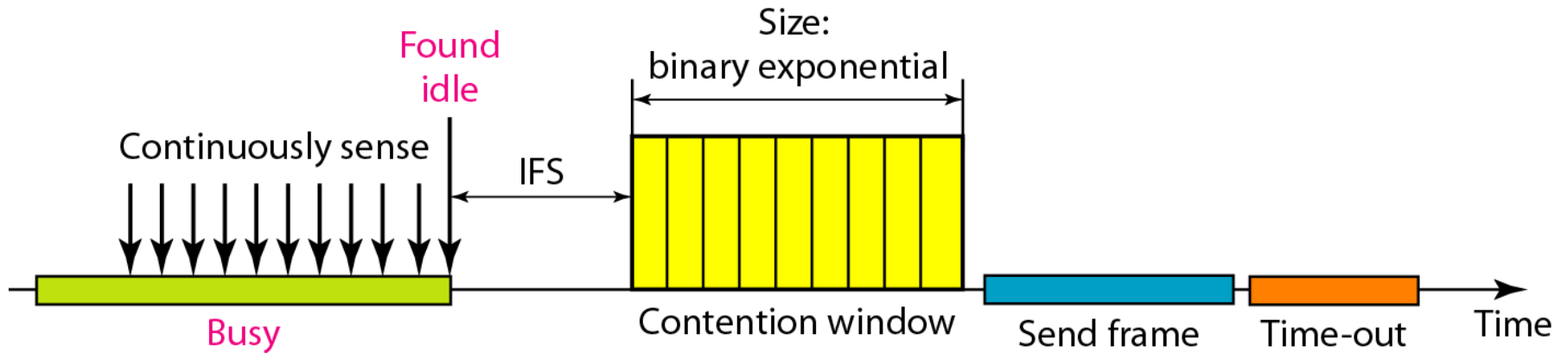
Key Aspects of CSMA/CA

- Collision Avoidance:** Nodes sense the channel, and if busy, they wait for a random backoff time before trying again.
- RTS/CTS Mechanism:** To solve the "hidden node problem," a station sends a **Request to Send (RTS)** and waits for a **Clear to Send (CTS)** from the receiver before sending data.
- ACK Required:** Every data frame is acknowledged by the receiver (ACK). If an ACK is not received, the frame is assumed lost and retransmitted.
- Binary Exponential Backoff:** If a transmission fails, the node waits for an increasing amount of time, known as backoff, to avoid further collisions.

CSMA/CA Process Flow

- 1.Sense Channel:** A station checks if the channel is busy.
- 2.Wait (if needed):** If busy, the station waits for an Interframe Space (IFS) and a random backoff time.
- 3.RTS/CTS:** Once idle, RTS/CTS frames are exchanged to reserve the channel.
- 4.Data Transmission:** The frame is transmitted.
- 5.ACK:** The receiver sends an acknowledgment.

Figure 16 *Timing in CSMA/CA*





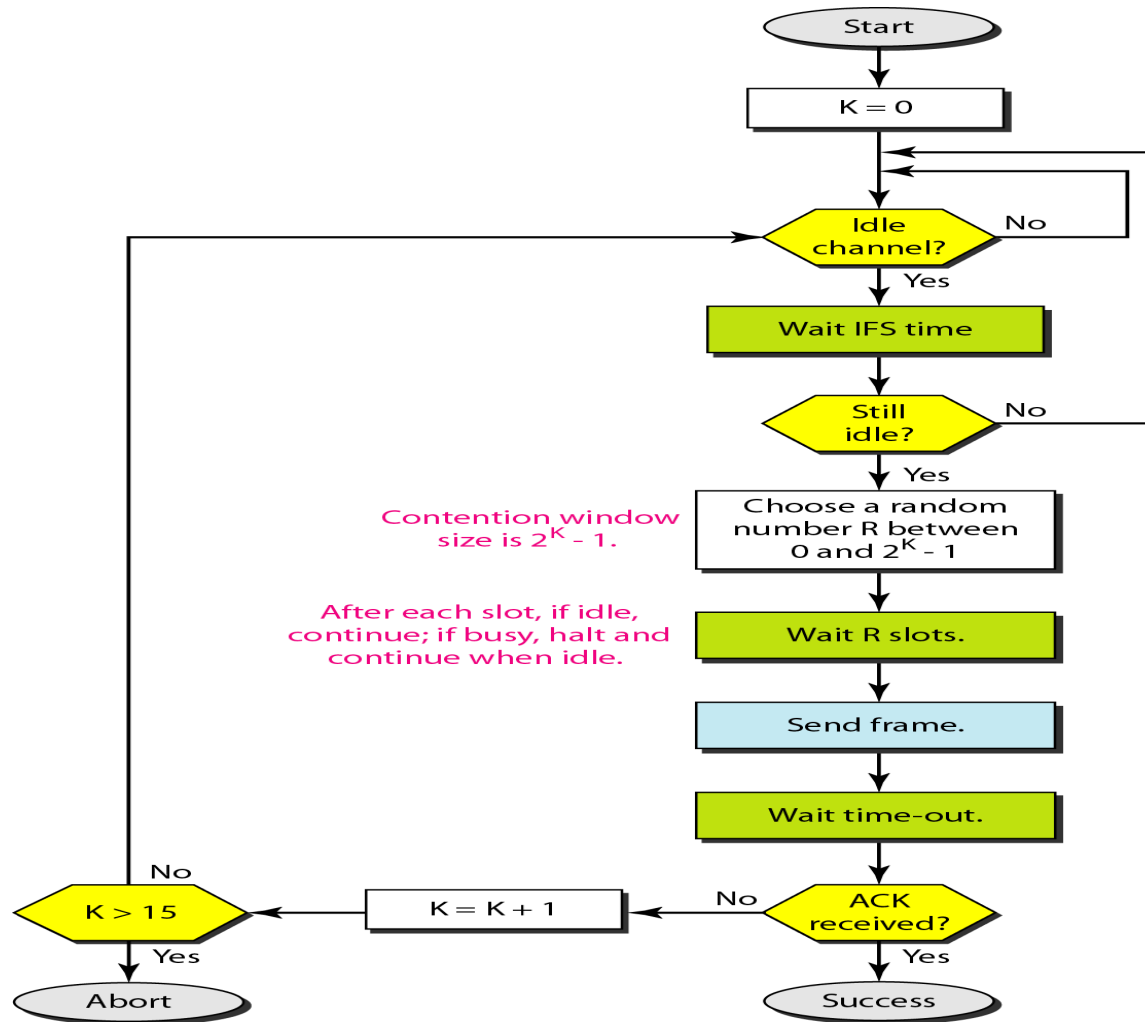
Note

In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Note

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

Figure 17 *Flow diagram for CSMA/CA*



CSMA	CSMA/CD	CSMA/CA
Carrier Sense Multiple Access	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Sense channel before sending	Sense and Detect collision while sending	Sense and Avoid collision before sending
Collisions may still happen	Collisions detected and transmission stops	Collisions prevented using extra techniques
Collision detection not supported	Collision detection supported	Collision detection not possible
Data corrupted if collision occurs	Sender stops, retransmits later	Collision minimized, rare
Mostly used in wired networks (basic)	Used in wired Ethernet networks	Used in wireless networks (Wi-Fi)
Used in early LAN systems	Ethernet (IEEE 802.3)	Wi-Fi (IEEE 802.11)
Medium efficiency	High efficiency in wired LAN	High efficiency in wireless LAN
Only carrier sensing technique	Collision detection + Jam signal	RTS/CTS + Random backoff
No jam signal	Jam signal used	No jam signal
Backoff optional	Backoff mandatory after collision	Backoff mandatory to avoid collision

ALOHA	CSMA
Pure random access protocol	Carrier sensing random access protocol
Station sends data anytime	Station first listens before sending
No checking of channel before transmission	Channel is sensed before transmission
Collisions are very frequent	Collisions are reduced
Collision detection not present	Collision avoidance by sensing
ACK is mandatory	ACK optional but used
Retransmission after random waiting time	Retransmission after sensing + backoff
Used in early wireless systems (satellite communication)	Used in LAN networks (Ethernet base)
Types: Pure ALOHA and Slotted ALOHA	Types: 1-persistent, non-persistent, p-persistent CSMA
Maximum throughput: Pure = 18%, Slotted = 36%	Higher throughput than ALOHA

CONTROLLED ACCESS

- *In **controlled access**, the stations consult one another to find which station has the right to send.*
- *A station cannot send unless it has been authorized by other stations.*
- *Unlike Random Access (ALOHA, CSMA), here transmission is organized, so collisions are avoided.*
- *Key Idea:*
“Permission is required before sending data.”

Controlled Access Categories

- **Reservation**
- **Polling**
- **Token Passing**

1. Reservation

- Reservation is a method where stations **reserve a time slot** before transmitting data.

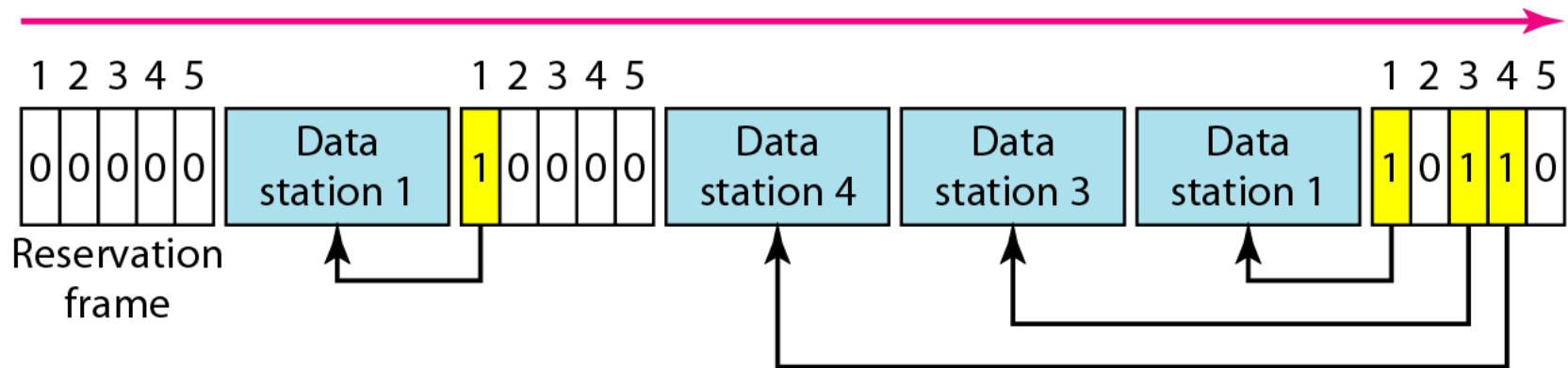
How it works:

- Time is divided into **cycles**
- Each cycle begins with a **reservation frame**
- Stations announce if they want to send data
- Slots are then assigned accordingly

Example

- Suppose 5 computers share a channel.
- Computer 2 and 4 want to send data
- They reserve slots in the reservation phase
- Then only those computers transmit in their assigned slots

Figure 18 *Reservation access method*



Reservation-Pros & Cons

Advantages

- Predictable access time and throughput.
- Priorities can be set for faster access.
- Supports QoS (Quality of Service) for applications like real-time audio/video.
- Efficient bandwidth utilization.

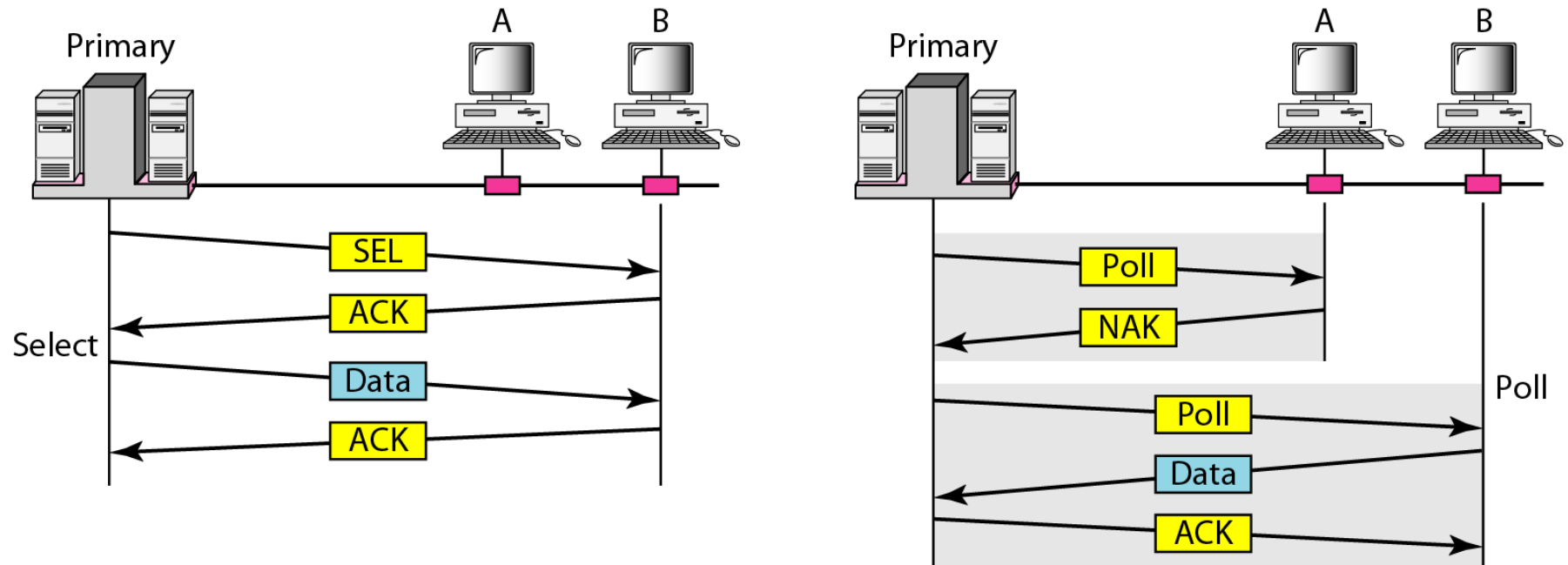
Disadvantages

- Performance decreases under light loads (wasted slots).
- High dependency on controlled synchronization.

2. Polling

- Here, a controller (primary station) polls each node (secondary station) in sequence, granting permission to send. The controller sends a message with the address of a station. The addressed station responds:
 - Sends data (if available)
 - Sends a NAK (poll reject) if no data is pending.

Figure 19 *Select and poll functions in polling access method*



Used in

- *Bluetooth networks*
- *Master-slave communication systems*

Polling- Pros & Cons

Advantages

- Predictable access times and bandwidth.
- High efficiency, since no slot is wasted.
- Priorities can be assigned.

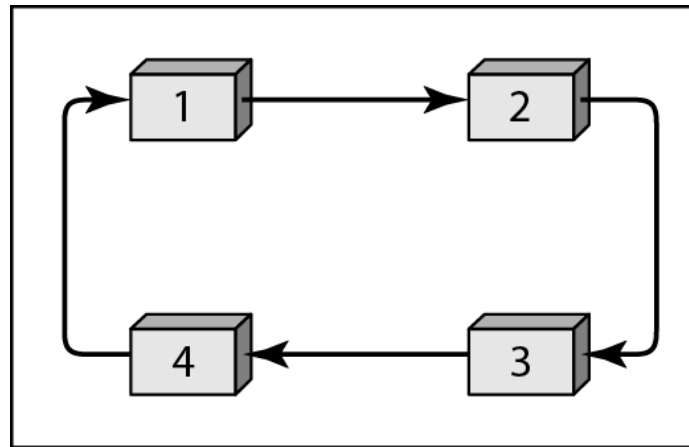
■ Disadvantages

- Extra overhead due to polling messages.
- High dependence on controller reliability.
- Turnaround time increases under light loads.

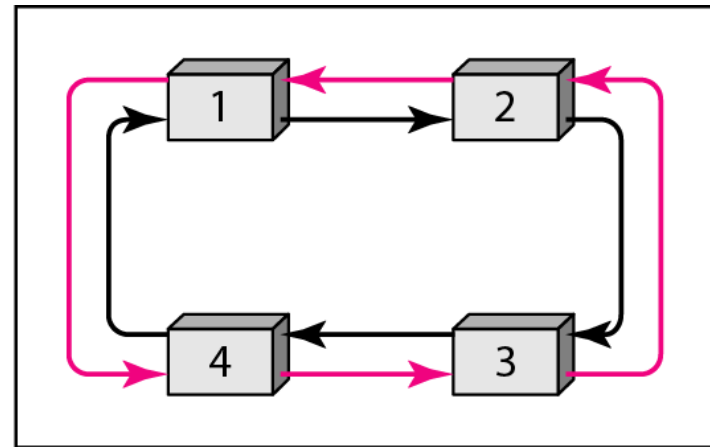
Token Passing

- In Token Passing, stations are logically connected (ring or bus). A special frame called a token circulates among stations:
- The token represents permission to transmit.
- A station can send data only when it holds the token.
- If no data is ready, the token is simply passed on.

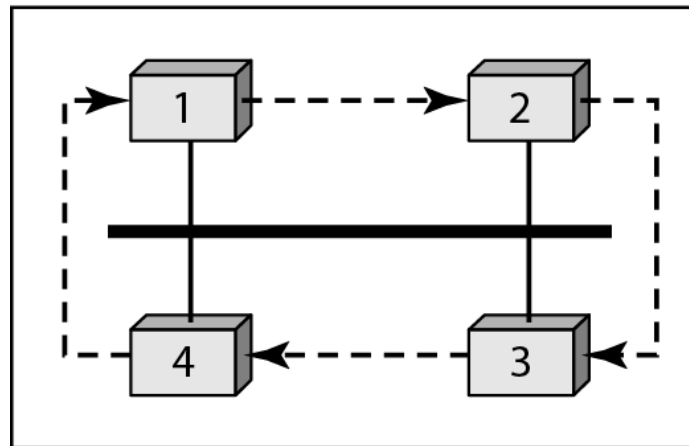
Figure 20 *Logical ring and physical topology in token-passing access method*



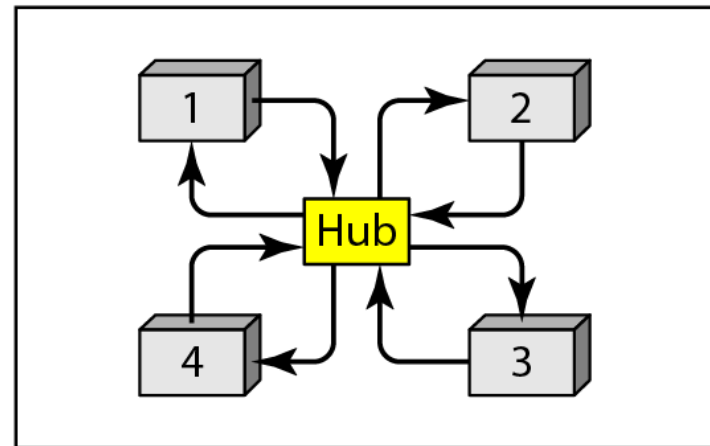
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Used in

1. Token Ring (IEEE 802.5)

2. FDDI networks

Token Passing- Pros & Cons

Advantages

- Excellent performance under high loads.
- Built-in error recovery and debugging features.
- Predictable throughput.

Disadvantages

- Expensive setup and hardware.
- Token management issues (lost/duplicate tokens, station failures).
- Complex implementation.

Reservation	Polling	Token Passing
Distributed control	Centralized control	Distributed control
No collision	No collision	No collision
Slot reservation gives permission	Primary asks stations	Token gives right to transmit
Reservation bits overhead	Polling messages overhead	Token circulation overhead
Low failure issue	Primary failure affects all	Token loss problem
Best for heavy traffic	Best for small networks	Best for ring/LAN networks

CHANNELIZATION

- ***Channelization** is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations..*
- *It is a Controlled Access Method (unlike Random Access)*

Topics discussed in this section:

Frequency-Division Multiple Access (FDMA)

Time-Division Multiple Access (TDMA)

Code-Division Multiple Access (CDMA)

1. FDMA (Frequency Division Multiple Access)

- The total bandwidth is divided into **different frequency bands**.
- Each user gets a separate frequency channel.

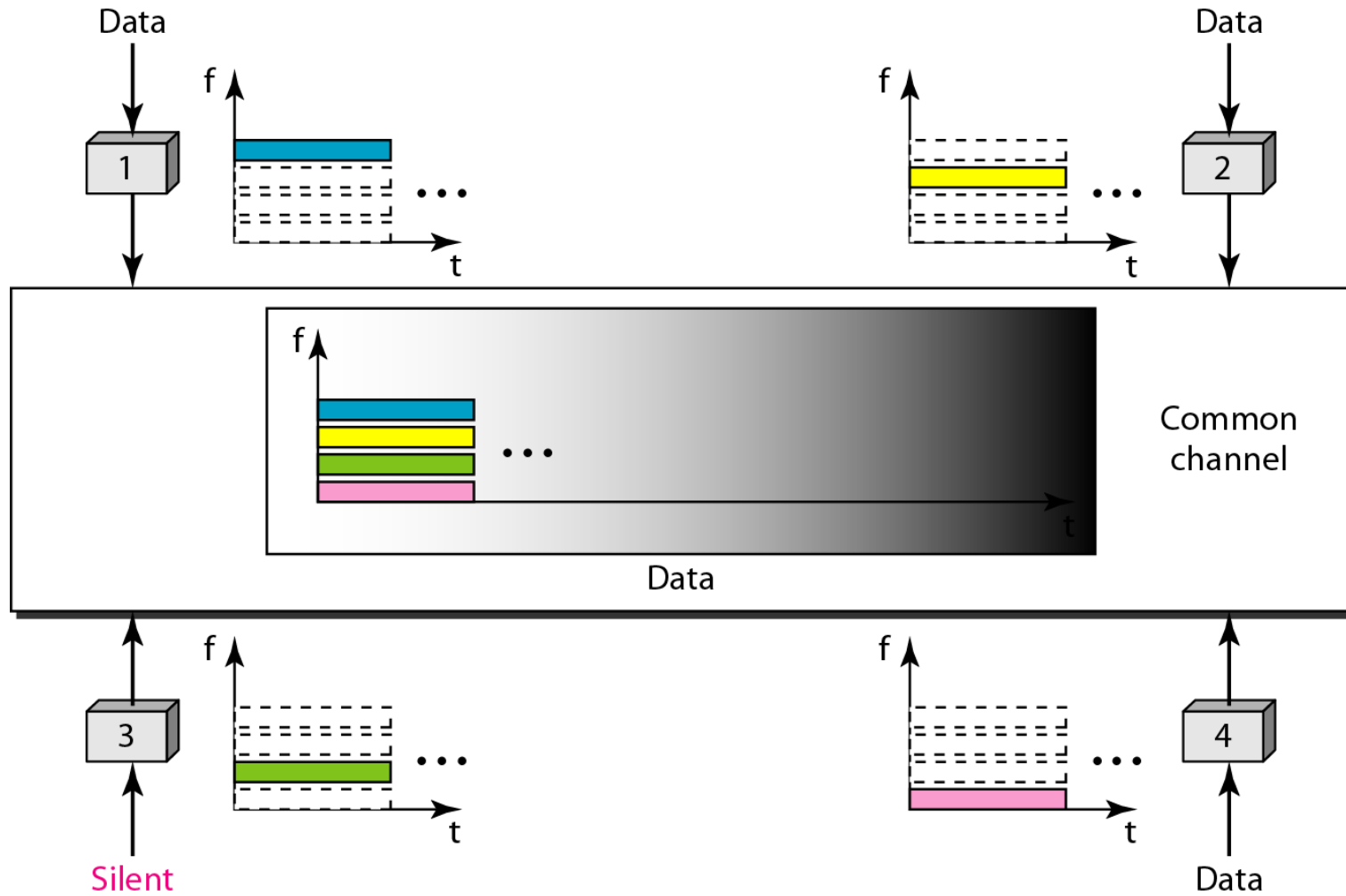
Working

- User 1 → Frequency Band f_1
- User 2 → Frequency Band f_2
- User 3 → Frequency Band f_3
- All users transmit at the same time, but on different frequencies.

Example

- Radio stations:
- FM 91.1
- FM 93.5
- FM 98.3
- Each operates at a different frequency.

Figure 21 *Frequency-division multiple access (FDMA)*



FDMA: Pros & Cons

Advantages

- No collision
- Simple technology
- Continuous transmission possible

Disadvantages

- Wastage of bandwidth (guard bands needed)
- Not efficient for bursty data



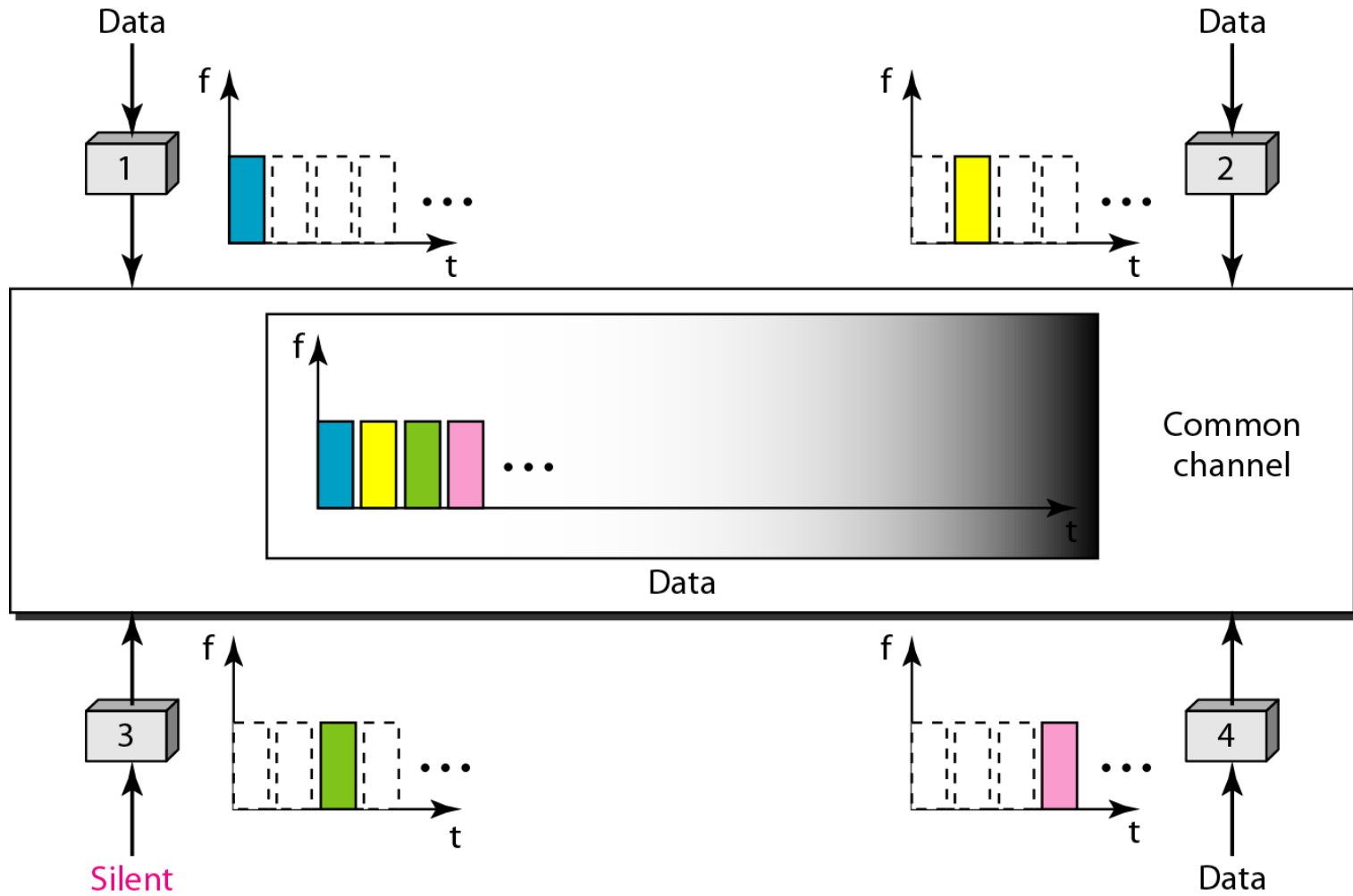
Note

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

2. TDMA (Time Division Multiple Access)

- The channel is divided into **time slots**.
- Each user gets the full bandwidth, but only for a specific time slot.
- Each station transmits only in its assigned slot.
- Needs synchronization.

Figure 22 *Time-division multiple access (TDMA)*



Note

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

CDMA

- **Code Division Multiple Access (CDMA)**
- In CDMA, all the stations can transmit data simultaneously.
- It allows each station to transmit data over the entire frequency all the time.
- Multiple simultaneous transmissions are separated by unique code sequence.
- Each user is assigned with a unique code sequence.
- Widely used in mobile communication.



Note

In CDMA, one channel carries all transmissions simultaneously.

Figure 23 *Simple idea of communication with code*

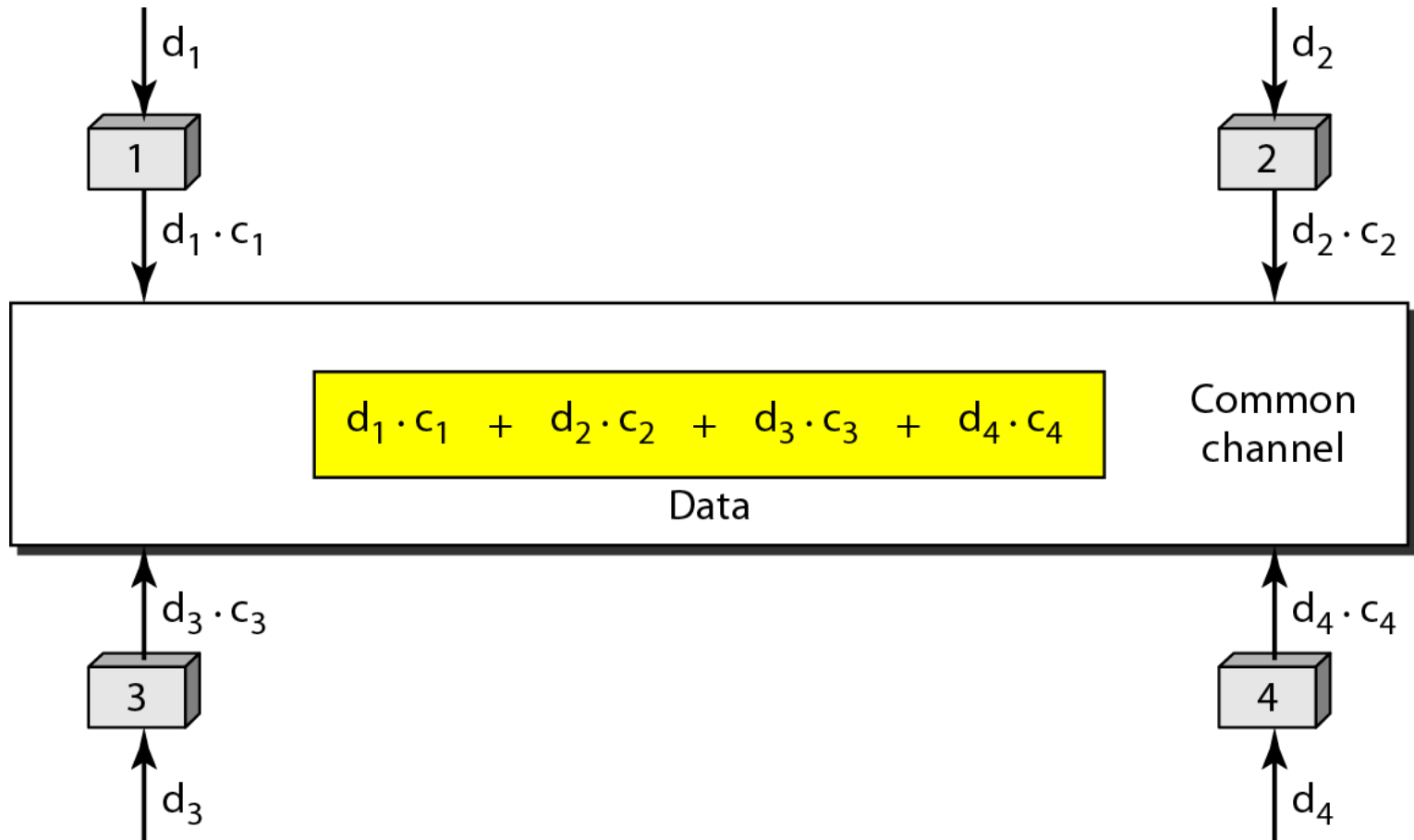


Figure 24 *Chip sequences*

C_1

[+1 +1 +1 +1]

C_2

[+1 -1 +1 -1]

C_3

[+1 +1 -1 -1]

C_4

[+1 -1 -1 +1]

Advantages of CDMA

- CDMA has a very high spectral capacity, supporting many users within a wide bandwidth.
- CDMA channels are hard to decode, improving cellular communication security.
- It provides better secure transmission capabilities.
- Dropouts only occur when the user is twice the distance from the base station.

Disadvantages of CDMA

- CDMA faces channel pollution when a user's phone connects to multiple cell sites, but only one has strong signal.
- CDMA requires time synchronization.
- Performance of the CDMA system decreases as the number of users increases.
- CDMA equipment tends to be more expensive due to its complexity.

Figure 25 *Data representation in CDMA*



Figure 26 *Sharing channel in CDMA*

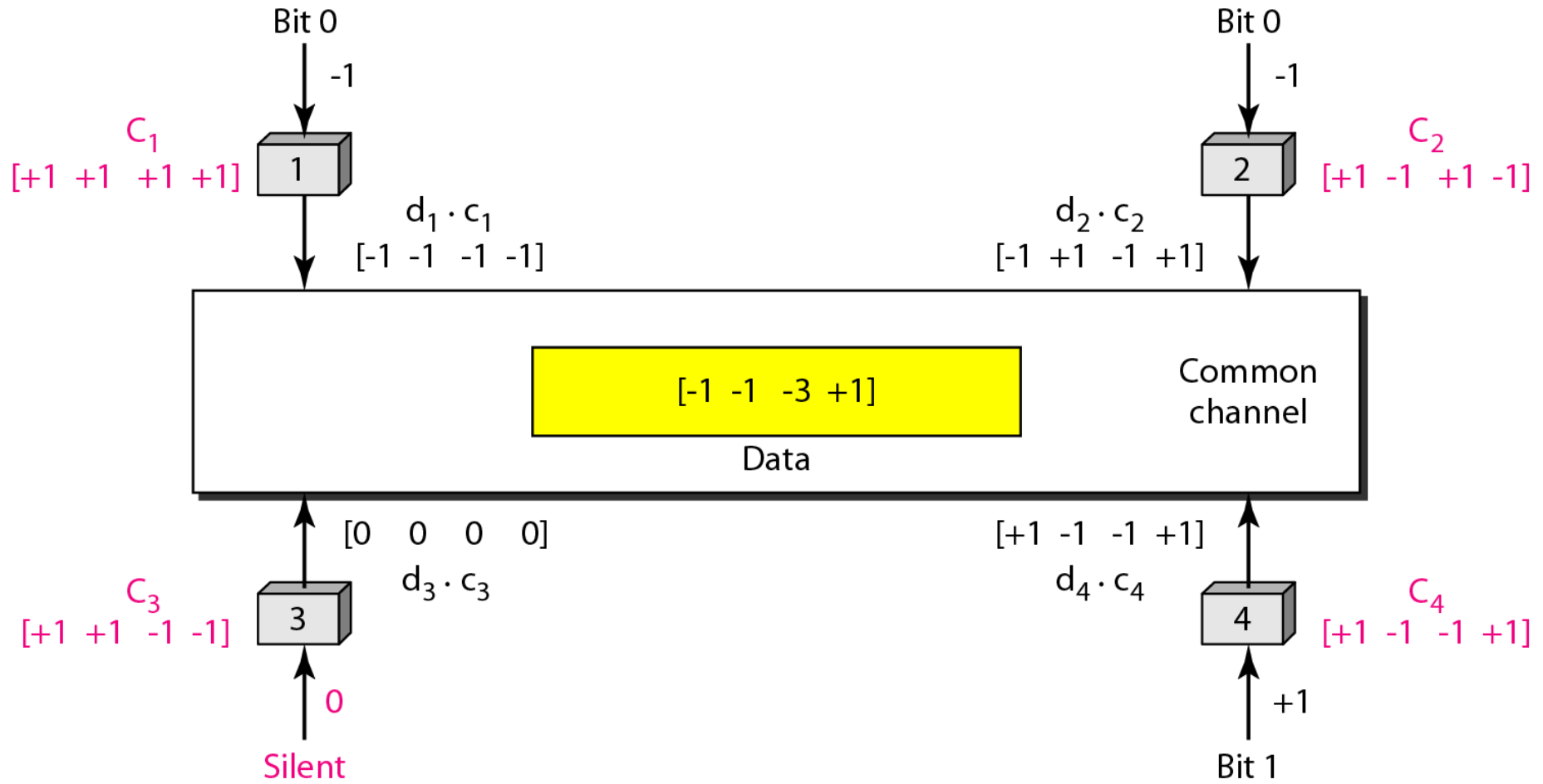


Figure 27 *Digital signal created by four stations in CDMA*

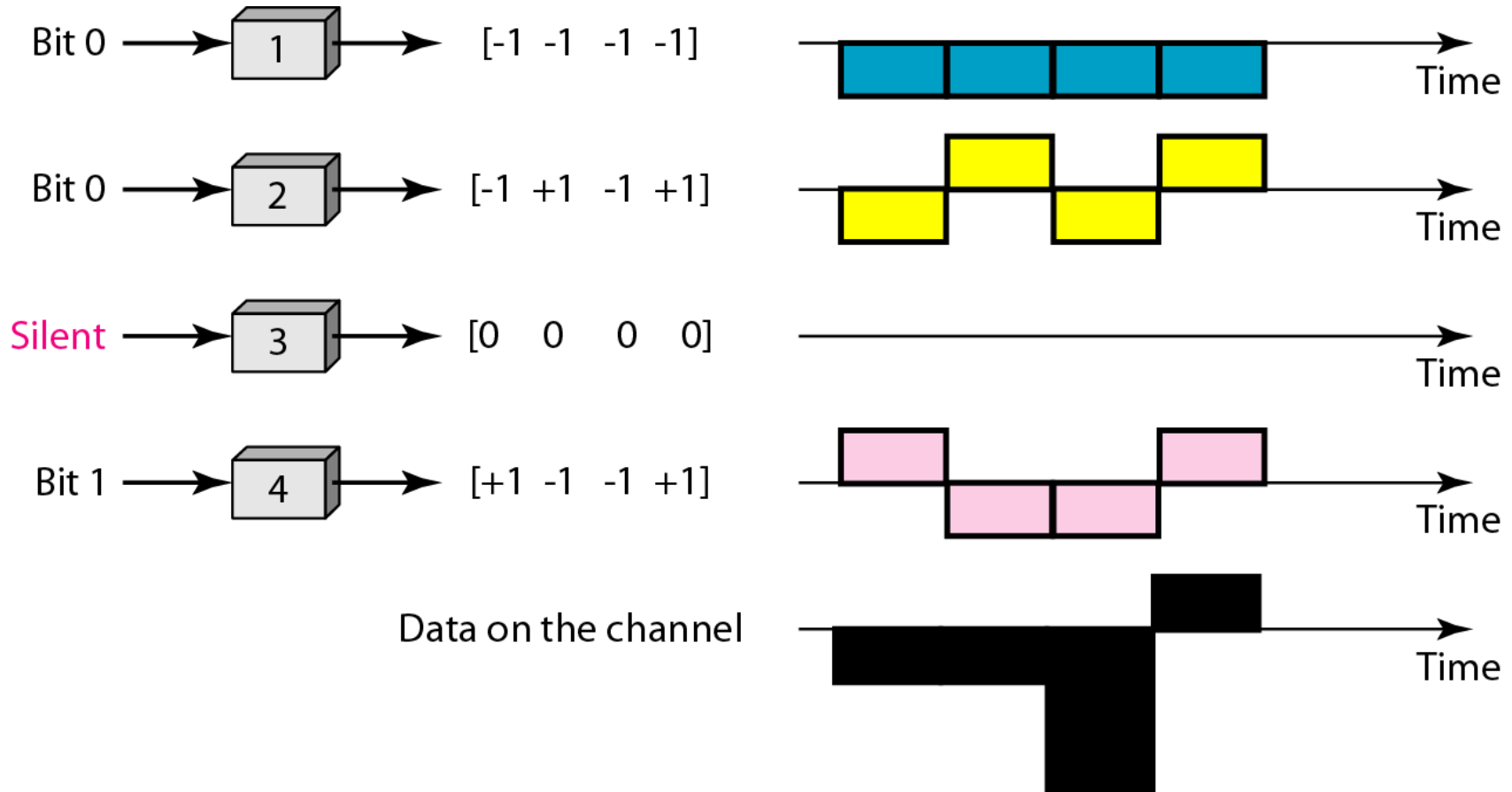


Figure 28 *Decoding of the composite signal for one in CDMA*

